

1. Project Name & Description

- **Project Name:** SkillHub - Online Courses Management Platform
- **Project Description:**
 - **What problem does it solve?** It solves the complexity of managing online learning, enabling users to browse, filter, and enroll in technical courses, while giving administrators total control over course catalogs and user management.
 - **Target Users:** Students/Learners looking for online courses, and Administrators/Instructors managing the platform.
 - **Main Purpose:** To provide a seamless RESTful web application for online education with full authentication, authorization, and profile customization.

2. Identify Users and Roles

Role Name	Permissions	Available Actions
Admin	Full system control	Add new courses, Edit/Delete courses, Change user roles (Make Admin/User), Delete user accounts, View enrolled courses statistics.
User (Student)	Access to student features	Register/Login, Browse and filter courses by title, category, and price, Enroll in courses, Edit profile details, Upload profile picture.

3. Define Main Features

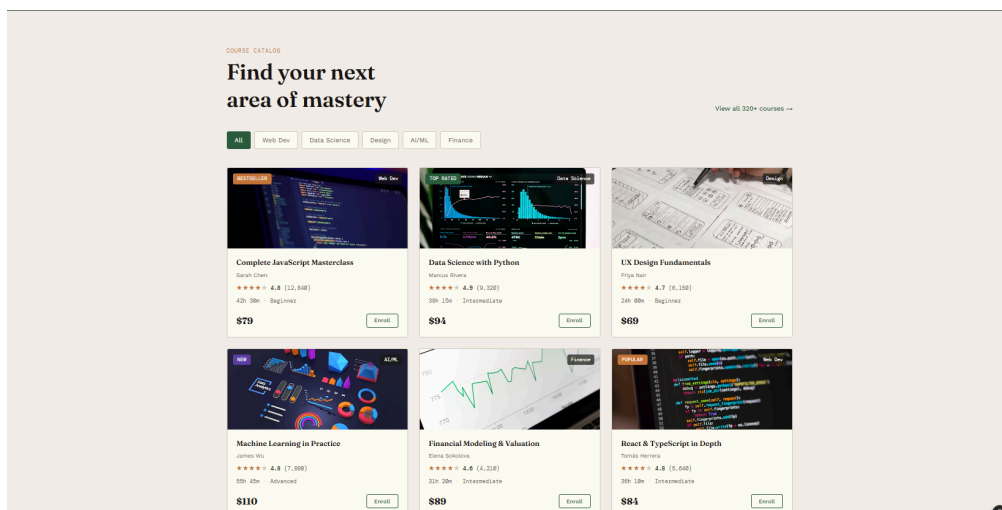
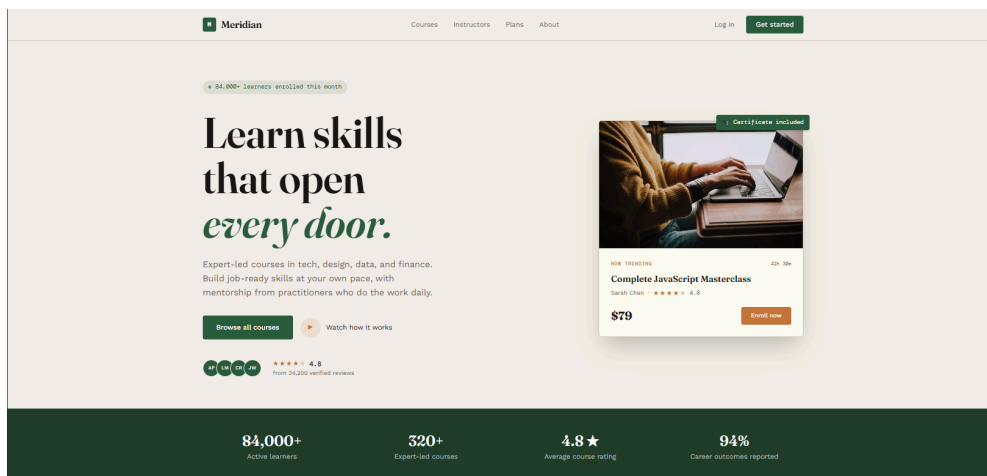
- **Authentication Features:**
 - User Registration (/signup)
 - User Login (/login)
 - Local Storage session management
- **Authorization Features:**
 - Role-based access control (Admin dashboard protection)
 - Protected routes based on user role (admin vs user)
- **CRUD Features (Courses & Users):**
 - **Courses Management:** Create (POST), Read (GET), Update (PUT), and Delete (DELETE) courses.
 - **Users Management:** Read users list, Update roles, and Delete users.
- **Image/File Upload Features:**
 - **User Profile Picture:**
 - Allowed format: Images (image/*)
 - Uploaded by: Users through device file selection (Base64 conversion).

4. Application Architecture & Tech Stack

Backend: Node.js & Express.js framework.

- **Architecture Pattern:** MVC Pattern (Models, Views/Frontend, Controllers, and Routes separation).
- **Data Storage:** JSON files (`courses-data.json` and `users-data.json`) handled via Node.js `fs` module.
- **Middleware:** `express.json()` for parsing incoming JSON bodies, custom static file serving.
- **REST API Endpoints Structure:**
 - `GET /api/v1/courses` - Get all courses
 - `POST /api/v1/courses` - Create a course (Admin)
 - `PUT /api/v1/courses/:id` - Update a course (Admin)
 - `DELETE /api/v1/courses/:id` - Delete a course (Admin)
 - `POST /api/v1/auth/signup` - Register user
 - `POST /api/v1/auth/login` - Login user
 - `GET /api/v1/auth/users` - Get all users (Admin)
 - `PATCH /api/v1/auth/users/:id/role` - Toggle user role (Admin)
 - `DELETE /api/v1/auth/users/:id` - Delete user account (Admin)
 - `POST /api/v1/auth/users/:id/enroll` - Enroll in a course (User)
 - `PATCH /api/v1/auth/users/:id/profile` - Update profile & avatar (User)

5. UI Design & Screens



Three steps to your next skill

01

Choose your course

Browse 320+ expert-led courses across tech, design, data, and finance. Filter by skill level, duration, and career outcome.

02

Learn at your pace

Watch HD video lectures, complete hands-on projects, and receive feedback from instructors who work in industry every day.

03

Earn your certificate

Complete the course and receive a verified certificate you can share directly to LinkedIn or add to your professional portfolio.

FEATURED INSTRUCTOR

Learn from practitioners, not just teachers.

Every Meridian instructor is an active practitioner — engineers at top-tier tech companies, designers at product-led startups, and analysts at leading financial firms.



Dr. Sarah Chen

Principal Engineer, Google • Stanford PhD

8

Courses

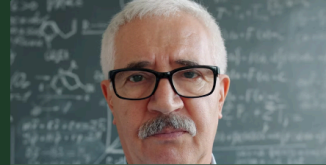
12,400

Students

4.9

★

Rating



"I much believe I genuinely believe that one great explanation, delivered clearly, can change someone's career trajectory entirely."

— Dr. Sarah Chen

STUDENT STORIES

Real outcomes, real voices

"The JavaScript Masterclass completely transformed how I write code. Within three months of finishing, I landed my dream role."



Aisha Patel

Frontend Engineer at Stripe

COMPLETE JAVASCRIPT MASTERCLASS

"Marcus's course is the most practical data science education I have found anywhere. Real datasets, real projects, career-changing skills."



Leon Müller

Data Analyst at Shopify

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Camille Rousseau

Product Designer at Linear

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